

Department of Computer Science

Semester Examination — Data Structures

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Subject:	Data Structures	Date:	20-07-2026
Max Marks:	100	Duration:	3 Hours

Q1. Explain Binary Search Trees. [10 Marks]

A Binary Search Tree (BST) is a hierarchical data structure in which each node has at most two children, referred to as the left and right child. For any given node, all elements in the left subtree are smaller than the node's value, and all elements in the right subtree are greater. This property makes searching, insertion, and deletion operations efficient, typically achieving $O(\log n)$ time complexity in a balanced tree. Traversal methods such as inorder, preorder, and postorder are commonly used to visit nodes in a BST, with inorder traversal producing values in sorted order.

Q2. Explain Time Complexity of Sorting Algorithms. [10 Marks]

Sorting algorithms are compared based on their time complexity in the best, average, and worst cases. Bubble Sort and Insertion Sort have a worst-case complexity of $O(n^2)$, making them inefficient for large datasets. Merge Sort and Quick Sort achieve an average case of $O(n \log n)$, with Merge Sort maintaining this bound even in the worst case due to its divide-and-conquer approach. Quick Sort, while faster in practice, can degrade to $O(n^2)$ in the worst case if pivot selection is poor.

Q3. Implement a Linked List (Theory). [20 Marks]

A linked list is a linear data structure where each element, called a node, contains data and a reference (pointer) to the next node in the sequence. Unlike arrays, linked lists do not require contiguous memory allocation, allowing efficient insertion and deletion without shifting elements. A singly linked list has nodes pointing only to the next node, while a doubly linked list maintains pointers to both the next and previous nodes, enabling traversal in both directions.

Q4. Graph Traversal Techniques. [30 Marks]

Graph traversal refers to the process of visiting all the nodes in a graph in a systematic manner. Breadth-First Search (BFS) explores neighbor nodes level by level using a queue, making it useful for finding the shortest path in unweighted graphs. Depth-First Search (DFS) explores as far as possible along each branch before backtracking, typically implemented using a stack or recursion. Both techniques are foundational for solving problems such as connectivity checks, cycle detection, and topological sorting.

Q5. Hashing and Collision Resolution. [30 Marks]

Hashing is a technique used to map data of arbitrary size to fixed-size values, known as hash codes, using a hash function. It enables average $O(1)$ time complexity for insertion, deletion, and lookup operations. Collisions occur when two different keys produce the same hash value.

Common collision resolution strategies include chaining, where each bucket maintains a linked list of entries, and open addressing, where the algorithm probes for the next available slot using techniques like linear probing, quadratic probing, or double hashing.

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